

## VINAY KUMAR CHAKALI

New York, NY (open to relocate)

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### SUMMARY

Full Stack Software Engineer with 1.5+ years of experience building production web applications with Python, React.js, TypeScript, and Node.js. Strong in Problem Solving, Data Structures and Algorithms, OOPs, and Low Level Design, with hands-on Multithreading, Concurrency, Microservices, and Distributed Systems experience. Integrates ChatGPT, Claude AI, LLM APIs, RAG, and LangChain into engineering workflows. Reduced API latency 30% and production bugs 35% through DBMS optimization and CI/CD discipline.

### WORK EXPERIENCE

**Starlite Infotech | Full Stack Software Engineer** | Bengaluru, India | Jan 2023 - Aug 2024

- Built full-stack features with **Python**, **React.js**, **TypeScript**, and **Node.js**, delivering **REST APIs** and responsive UI components structured with **OOPs** and **Design Patterns** across core production modules.
- Diagnosed slow reporting endpoints, so refactored **PostgreSQL** and **MySQL** queries with composite indexing and connection pooling using **LLD** and **DBMS** internals, cutting average API latency by 30% across 12 endpoints.
- Resolved race conditions in parallel request handling with **Multithreading** and **Concurrency** controls in backend services, improving throughput by 40% within a **Microservices** architecture.
- Cut production bug rate by 35% by building **PyTest** and **Jest** suites into **CI/CD** pipelines with code review gates, enforcing test coverage thresholds before every production release.
- Containerized services with **Docker** and **Kubernetes**, applying **HLD** and **Distributed Systems** principles to improve deployment consistency and cut page load times by 20%.

### PROJECT EXPERIENCE

**Polish Music Matching Platform | Full Stack Software Engineer** | Jun 2025 | **[GitHub]** | **[Live]**

- Designed a real-time, high-**Concurrency** music matching platform with a 30-second active-window matching engine using **FastAPI** and **WebSockets**, structuring the system with **HLD** diagrams before implementation.
- Optimized the matching algorithm with efficient **Data Structures** (hash-indexed active windows) and async event handling, achieving match detection and live messaging in under 4 seconds under concurrent load.
- Secured the platform with **JWT** authentication, bcrypt hashing, and dual-layer geolocation **REST APIs**, backed by a unit and integration test suite covering authentication and **WebSocket** flows before cloud deployment.

**Book Data Pipeline | Backend Software Engineer** | Jun 2025 | **[GitHub]**

- Engineered a distributed data ingestion pipeline with **Python**, **Requests**, and **BeautifulSoup** using **OOPs** class design, automating structured catalog extraction and reducing manual gathering time by 98%.
- Architected a normalized **DBMS** schema with **UNIQUE** constraints and idempotent deduplication logic, applying **LLD** principles across **SQLite** with a documented migration path to **PostgreSQL** and **MongoDB** for unstructured metadata.
- Exposed data via **FastAPI REST API** endpoints with parameterized queries, server-side pagination, and injection-safe fuzzy search, load-tested for consistent sub-100ms query response.

### TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, SQL, Java, HTML5, CSS3
- **Frameworks and Libraries:** React.js, Node.js, Express.js, FastAPI, SQLAlchemy, Pydantic
- **Databases and DBMS:** PostgreSQL, MySQL, MongoDB, SQLite, relational database design, indexing, query optimization
- **Core CS:** Problem Solving, Data Structures and Algorithms (DSA), OOPs, Computer Networks, Operating Systems, Low Level Design (LLD), High Level Design (HLD), Design Patterns, Multithreading, Concurrency, Distributed Systems, Microservices, Kafka (event streaming fundamentals)
- **AI Engineering:** ChatGPT, Claude AI, LLM APIs (OpenAI, Anthropic), RAG, LangChain, LangGraph, AI Agents
- **Tools and Cloud:** GitHub, Git, Docker, Kubernetes, CI/CD, GitHub Actions, Postman, AWS, Azure

### EDUCATION

**Quinnipiac University | Master of Science - Computer Science** | Hamden, CT | May 2026

- **Relevant Coursework:** Data Structures and Algorithms, Distributed Operating Systems, Database Systems, Computer Networks, Software System Design, Deep Learning, Cloud Computing and Microservices.
- **Certifications:** LangChain and RAG Systems (Deep Learning.AI) | Docker and Kubernetes for Developers (KodeKloud).